
Trace–Answer Compatibility Emerges at Depth and Mediates Prediction in Diffusion Language Models

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Abstract

1 Diffusion language models expose latent correctness signals in their hidden states,
2 but it is unclear what those signals represent or whether the model uses them.
3 Studying LLaDA-8B and Dream-7B with autoregressive controls (Qwen3-8B and
4 Llama-3.1-8B), we identify a concrete representational object: late hidden states
5 encode the *compatibility* between a generated trace and its final answer. A 2×2
6 intervention crosses the non-answer trace from correct- and wrong-answer genera-
7 tions with correct and wrong final answers, isolating a trace–answer interaction.
8 Without using correctness labels, compatibility is linearly decodable from deep
9 states at AUC 0.919 in LLaDA and 0.897 in Dream and transfers across equivalent
10 answer formats, while the shallow geometry is weak and poorly aligned. Crucially,
11 this representation is causally predictive: re-masking the answer and patching deep
12 answer-slot activations between correct- and wrong-source traces mediates 66%
13 and 80% of the trace-conditioned answer-likelihood gap, with matched pre-answer
14 controls that are null or small. The effect is dose-dependent, requires a question-
15 matched donor, and is sensitive to controlled numerical and ordering corruptions
16 of the trace. Objective-matched AR controls show that compatibility is not unique
17 to diffusion models; in the diffusion models studied, however, it forms later and
18 becomes more causally concentrated at masked-answer slots.

19 1 Introduction

20 Diffusion language models generate by iteratively reconstructing masked tokens rather than predicting
21 strictly left-to-right [Kishore et al., 2023, Yu et al., 2025b], which changes the objects available to
22 interpretability: no fixed position summarizes a prefix, and token confidence need not track solution
23 quality. Recent work nevertheless finds that linear readouts [Alain and Bengio, 2018] can decode
24 whether diffusion-model generations are correct [Yadav et al., 2026]. What such a readout represents,
25 when it forms, and whether the model uses it remain unresolved. A high correctness AUROC does
26 not answer those questions: it can arise from question difficulty, the lexical form of the answer, a
27 property of the trace, or compatibility between the trace and answer, and these sources imply different
28 internal representations. We study this ambiguity in two diffusion families, the natively trained
29 LLaDA-8B [Nie et al., 2025] and the AR-initialized Dream-7B [Ye et al., 2025], using Qwen3-8B and
30 Llama-3.1-8B as reference controls rather than leaderboard competitors, following one evidence chain
31 from measurement control to causal localization. Evaluating only correct–incorrect pairs for the same
32 question, against answer-only and first-block floors, both diffusion models acquire a task-transferable
33 correctness direction primarily at depth: GSM8K-to-MATH AUC rises from 0.517 to 0.736 in LLaDA
34 and 0.495 to 0.666 in Dream, whereas both AR controls transfer from their first block. This is a
35 difference in representational formation, not evidence about where reasoning is computed. Crossing
36 the non-answer trace from correct- and wrong-answer generations with correct and wrong final
37 answers isolates a trace–answer interaction. Discarding the original correctness labels, we decode

the controlled compatibility factor directly: deep all-position compatibility AUC reaches 0.919 in LLaDA and 0.897 in Dream and transfers across equivalent answer formats, while the shallow geometry is weak or unaligned. Inserting the same correct answer into correct- and wrong-source traces, re-masking it, and patching the hidden states that predict it [Heimersheim and Nanda, 2024], deep correct-trace states rescue answer likelihood in the wrong trace and reverse patches damage it; the symmetric intervention mediates 66% of LLaDA’s and 80% of Dream’s trace-conditioned likelihood gap, with matched pre-answer controls null or small. The effect is dose-dependent, requires a question-matched donor, and tracks controlled numerical and ordering corruption of the trace. Objective-matched AR controls also show deep mediation but retain large influence from preceding positions, so the diffusion distinction is late causal concentration at masked-answer slots, not exclusive possession of compatibility.

Our contributions are: (1) We resolve latent diffusion-LM correctness readouts into a replicated, late-forming trace–answer compatibility geometry that is decodable *without* correctness labels, using within-question, answer-format, cross-task, and factorial controls. (2) Via objective-aligned activation patching with dose, donor, and location controls, we show these late answer-prediction states causally mediate trace-conditioned answer preference and track controlled numerical and ordering corruption of the trace. (3) Objective-matched AR controls delimit the claim: compatibility is not diffusion-unique, but its formation depth and its concentration at masked-answer slots differ. We do not claim that diffusion LMs are intrinsically more interpretable, that trace source is a process-validity label, or that a single head computes compatibility. The source-token paths and features that construct the representation, and the training mechanism behind it, remain open.

2 Related Work

Linear probes are widely used to test whether information is accessible in intermediate representations [Alain and Bengio, 2018]. Their simplicity is useful, but a successful probe does not establish that the model uses the decoded feature, that the feature was computed at the probed layer, or that the representation has abstracted away from the input [Hewitt and Liang, 2019, Belinkov, 2021]. Control tasks and selectivity measurements address the capacity of the probe. Our concern is orthogonal: even a low-capacity readout can achieve high accuracy by exploiting information that is genuinely present but irrelevant to the intended interpretation. We therefore compare the deep readout with text-only and first-block floors.

Recent work trains probes or lightweight verifiers over language-model activations to rank candidate reasoning trajectories. TrajSelector, CLUE, LiLaVe, and DUCHESS report that latent readouts can improve best-of- N selection over token confidence or self-consistency [Yu et al., 2025a, Liang et al., 2025, Piotrowski et al., 2025, Jiang et al., 2025]. These methods establish the practical value of representation-based reranking. They also make the evaluation unit important: a selector must rank different attempts to the *same* problem. Pooled AUROC includes many cross-problem pairs that the selector never encounters. Our within-question AUC directly measures the required ordering. Several reported results illustrate why the distinction matters. A readout can have high pooled AUROC while failing to improve best-of- N selection, whereas another setting with comparable pooled performance can produce a useful selector. We do not propose a new selector. Instead, we show empirically that within-question AUC predicts selection gain better than pooled AUROC and can therefore diagnose such discrepancies.

Diffusion language models generate by iteratively denoising masked tokens under a discrete or masked diffusion objective [Kishore et al., 2023, Yu et al., 2025b]; we study two instances, the natively trained LLaDA [Nie et al., 2025] and the AR-initialized Dream [Ye et al., 2025]. Yadav et al. [2026] compare latent correctness readouts with model confidence in LLaDA and Dream, including AR controls. Their experiments show that correctness can be decoded from diffusion-model hidden states even when token confidence is weak. Their primary protocol uses one generation per prompt and consequently reports pooled AUROC. They also report a strong prompt-only control, indicating that question difficulty explains a substantial component of the score. We view that result as complementary rather than contradictory. We instantiate a closely matched protocol in our own harness and recover the same qualitative separation. Multiple generations per question then let us ask a different question: whether the readout can distinguish a correct attempt from an incorrect attempt to the same prompt. We further ask how much of that signal is already present in the output string or first transformer block. Our main extension is interpretive rather than metric-only: we cross trace and

Model	Class	Blocks	Hidden size
Qwen3-8B	autoregressive	36	4096
Llama-3.1-8B-Instruct	autoregressive	32	4096
LLaDA-8B-Instruct	masked diffusion	32	4096
Dream-7B-Instruct	AR-initialized diffusion	28	3584

Table 1: Models in the main comparison.

answer factors, decode compatibility without correctness labels, and intervene on the model’s own conditional answer distribution.

Most language-model interpretability tools were developed for causal transformers. Applying the same layer, token position, or logit-based lens to a bidirectional masked model assumes that these objects have comparable roles. This assumption need not hold. Our objective-aligned answer score is a simple example: an AR model naturally scores answer tokens left-to-right, while a diffusion model can score an answer after re-masking it in the context of the completed solution. We use this comparison not to claim a mechanism, but to motivate architecture-aware controls when interpreting readouts. Causal mediation analysis and activation patching have localized where information is causally used in autoregressive transformers [Vig et al., 2020, Meng et al., 2023]; we port these objective-aligned interventions to diffusion models and find the causal contribution concentrated later, at masked-answer slots. Controlled input recombination and activation patching [Heimersheim and Nanda, 2024] play complementary roles in our analysis. Crossing trace source and answer identity identifies which relation a readout responds to; bidirectional state transplants then test whether that relation affects prediction. Component and held-out head patches refine the causal localization, while their asymmetry and non-additivity prevent us from treating them as a complete circuit decomposition.

3 Experimental Setup

Models and data. Our primary comparison uses the natively trained masked-diffusion model LLaDA-8B-Instruct and the AR-initialized diffusion model Dream-7B, with Qwen3-8B and Llama-3.1-8B-Instruct as matched-scale autoregressive controls (Table 1). GSM8K is the primary task; MATH provides a second numeric-reasoning task. The main layer artifacts use $n = 150$ questions with $K = 6$ independently sampled generations per model at temperature 0.9. Diffusion decoding uses 128 denoising steps over a 256-token region; correctness is exact numeric match. Dream requires a one-position logit shift inherited from its AR initialization, applied consistently during generation and scoring.

Measurement. Because pooled AUROC conflates question difficulty with per-attempt correctness, we report *within-question* AUC: only correct–incorrect pairs from the *same* question contribute, with question-level cross-validation and question bootstraps (Appendix A justifies this target). We control two floors: an *answer-only* baseline (seven features of the parsed answer string, no internals) and a *first-block* readout (the output of transformer block 0, not the raw embedding). For each generation we also compute a training-free *re-masked conditional answer likelihood*: we mask the detected answer span and score the mean log-probability the diffusion model assigns to it given the rest of the solution (for AR models, the teacher-forced answer-token log-probability). The default probe is L2-regularized logistic regression over standardized hidden states, saved for every second block. All causal and probe results use the question as the independent unit; bootstraps keep a question’s variants together.

4 A Task-General Correctness Readout Forms at Depth

We first locate *when* a correctness-associated direction becomes linearly accessible. Using all-position within-question AUC at fixed depth endpoints on GSM8K (Table 2), LLaDA rises from 0.451 at block 0 to 0.826 at block 30 and Dream from 0.556 to 0.800, whereas Qwen starts at a high shallow floor (0.820) and gains little. A per-cell layer-by-position analysis (Fig. 1) shows the AR floor is a shallow terminal-position stripe, while the diffusion signal emerges in deep blocks across several positions. The depth dissociation replicates under an independent generation seed:

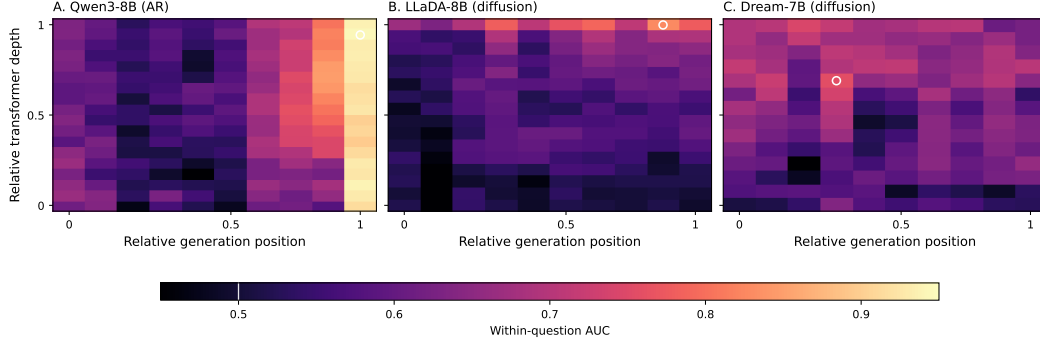


Figure 1: Layer-by-position within-question AUC. Qwen shows a terminal-position stripe from shallow depth; LLaDA’s signal emerges in deep blocks across several positions, while Dream is weaker and less terminally localized. Open circles mark each model’s highest-AUC cell; all panels share one color scale.

Model	First block	Deepest sampled	Depth gain	95% CI
Qwen3-8B	0.820	0.869	+0.049	[+0.005, +0.094]
LLaDA-8B	0.451	0.826	+0.375	[+0.297, +0.455]
Dream-7B	0.556	0.800	+0.244	[+0.164, +0.328]

Table 2: GSM8K all-position within-question AUC across fixed depth endpoints. Intervals bootstrap questions using fixed out-of-fold scores.

LLaDA gains +0.297 [0.219, 0.377] from the first to deepest sampled block, whereas Qwen gains +0.013 [−0.024, 0.054], a depth-gain contrast of +0.284 [0.193, 0.370].

This shallow-floor-versus-late-gain split is not an in-domain artifact. We train the standardized readout on one task and evaluate its within-question AUC on the other, holding the coordinate system fixed (Table 3). Both AR models transfer substantially from the first block, while the diffusion models begin at chance and become transferable only in later blocks: LLaDA reaches 0.736 and Dream 0.666 at their deepest blocks. The paired LLaDA-versus-Qwen depth-gain contrast is +0.202 [0.086, 0.319] toward MATH and +0.198 [0.088, 0.314] in reverse.

Two scope notes. The depth-gain contrast is strong on GSM8K but same-signed and inconclusive on MATH at $n = 150$, so we scope the dissociation to GSM8K. And single-layer ablations do *not* reproduce the decodability curves: accessibility and causal necessity are distinct measurements (Appendix F), which is why the causal test in Section 6 intervenes directly at the answer-prediction positions rather than ablating whole layers.

5 What Does the Correctness Readout Read?

To identify *what* the deep readout responds to, we run a 2×2 trace–answer intervention. For each question with a correct-answer and a wrong-answer generation, we split each into its parsed final numeric answer and the remaining non-answer trace, giving a correct-source trace R_c , a wrong-source trace R_w , the correct answer A_c , and a sampled wrong answer A_w . Crossing the two factors yields four variants $s(R, A)$; only the final numeric span is swapped, and every rendering is checked for numeric equivalence. We use “trace source” rather than “correct reasoning”: matching the final answer does not certify each intermediate step. Cohorts are 63 Qwen, 79 Llama, 114 LLaDA, and 101 Dream questions; probes are trained only on the original 150-question artifacts and each variant is scored by the fold that excluded its question.

The four scores define three paired contrasts,

$$\begin{aligned} E_{\text{answer}} &= \frac{1}{2}[(s_{cc} - s_{cw}) + (s_{wc} - s_{ww})], & E_{\text{trace}} &= \frac{1}{2}[(s_{cc} - s_{wc}) + (s_{cw} - s_{ww})], \\ E_{\text{int}} &= \frac{1}{2}[(s_{cc} - s_{cw}) - (s_{wc} - s_{ww})], \end{aligned} \quad (1)$$

Model	First	Deepest	Transfer depth gain [95% CI]
Qwen3-8B	0.754	0.772	+0.017 [−0.037, 0.071]
Llama-3.1-8B	0.712	0.787	+0.074 [0.009, 0.141]
LLaDA-8B	0.517	0.736	+0.219 [0.115, 0.322]
Dream-7B	0.495	0.666	+0.170 [0.079, 0.265]

Table 3: Cross-task within-question AUC (GSM8K→MATH). Probes are trained on GSM8K and applied directly to MATH; intervals bootstrap target questions with fixed source-trained probes. The reverse direction is same-signed (Appendix G).

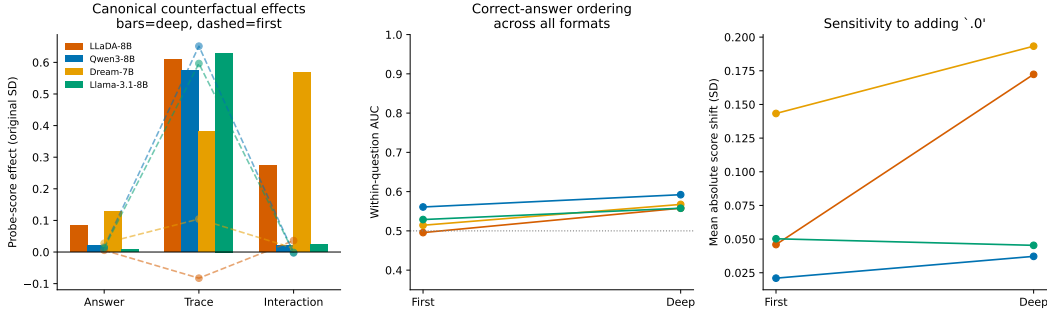


Figure 2: Counterfactual readout controls. **A**: all-position answer, trace, and interaction effects (deepest block; dashed = first block). **B**: within-question ordering of correct- versus wrong-answer counterfactuals across three answer renderings. **C**: residual answer-format sensitivity from appending “.0”.

160 reported in source-score standard deviations (not AUCs). Under all-position pooling the AR readout
161 already exposes a large trace effect at the first block (≈ 0.6 SD for Qwen and Llama), whereas
162 LLaDA and Dream have no reliable first-block trace effect but reach +0.608 and +0.382 SD at depth
163 (all-position table in Appendix G).

164 The interaction is clearest at the exact answer span (Table 4). Inserting the correct answer *raises* the
165 score inside a correct-source trace but *lowers* it inside a wrong-source trace: for LLaDA, +1.406
166 SD inside R_c versus -0.476 inside R_w ; Dream shows the same reversal more strongly. A deep
167 diffusion answer state therefore behaves not as an answer-identity lookup but as a relation between
168 the answer and its surrounding trace. The relation is present in deep Llama too; the architecture
169 result is its *emergence across depth* (deep-minus-first answer-span interaction +0.193 Qwen, +0.689
170 Llama, +0.891 LLaDA, +1.519 Dream), with matched diffusion-minus-AR emergence contrasts
171 +0.698 [0.505, 0.885] and +0.830 [0.525, 1.124].

Model	$\Delta_A(R_c)$, deep/answer	$\Delta_A(R_w)$, deep/answer	E_{int} , deep/answer
Qwen3-8B	+0.535 [0.348, 0.722]	+0.133 [−0.033, 0.298]	+0.201 [0.092, 0.307]
Llama-3.1-8B	+1.021 [0.816, 1.235]	−0.399 [−0.616, −0.169]	+0.710 [0.557, 0.871]
LLaDA-8B	+1.406 [1.205, 1.606]	−0.476 [−0.705, −0.246]	+0.941 [0.807, 1.069]
Dream-7B	+1.916 [1.651, 2.181]	−1.037 [−1.412, −0.641]	+1.477 [1.214, 1.737]

Table 4: Conditional answer effects at the deepest sampled block and exact answer positions. The correct answer is favored inside R_c and can be disfavored inside R_w .

172 Answer format is real but secondary: equivalent formatting shifts exact-answer scores (especially
173 leading zeros), yet the trace effects and first-versus-deep pattern persist under all three renderings,
174 and removing the answer-format subspace leaves the deep LLaDA and Dream readouts unchanged
175 (Appendix G). These interventions identify what the trained probe responds to under controlled input
176 recombination; they do not yet show the model uses this direction, which Section 6 tests causally.

6 A Causal Compatibility Representation in Diffusion LMs

We first remove the original correctness labels from the representational test, then ask whether the identified deep states causally affect the model’s own answer distribution.

Compatibility is directly encoded in deep states. We train question-grouped decoders for compatibility—distinguishing the original pairings (R_c, A_c) and (R_w, A_w) from the crossed pairings (R_c, A_w) and (R_w, A_c) —without using correctness labels. All-position compatibility AUC rises from 0.501 to 0.919 in LLaDA and from 0.547 to 0.897 in Dream. Canonical-to-other-format transfer reaches 0.887 and 0.886, while mean cross-question contrast alignment rises from 0.009 to 0.390 in LLaDA and from 0.061 to 0.415 in Dream (Table 5). Alignment is the mean cosine between per-question compatibility contrast directions: values near zero mean each question uses an idiosyncratic direction, whereas larger values indicate a shared, question-general compatibility axis. Deep AR controls also encode compatibility, so compatibility is not diffusion-unique. In the diffusion models studied, the distinctive result is its organization across depth from a weak or poorly aligned shallow geometry.

Model	State	Compatibility AUC	Cross-format AUC	Contrast alignment
LLaDA-8B	first/all	0.501	0.499	0.009
LLaDA-8B	deep/all	0.919	0.887	0.390
Dream-7B	first/all	0.547	0.508	0.061
Dream-7B	deep/all	0.897	0.886	0.415

Table 5: Label-free compatibility geometry. Cross-format decoders are trained on canonical answers and evaluated on leading-zero and decimal-equivalent answers; alignment is mean pairwise cosine between question-level contrast vectors.

Late answer states causally mediate trace-conditioned prediction. For each crossed question we build two sequences with the same correct answer, one in R_c and one in R_w , re-mask the answer, and measure its conditional log-likelihood. At each block we patch the masked answer-prediction states between traces (a forward *rescue* inserts R_c states into R_w ; the reverse *damage* does the opposite), with a matched pre-answer span as location control. The unpatched correct answer is much more likely under R_c than R_w (a 4.118-nat gap in LLaDA, 6.471 in Dream). Shallow patches are null; the effect emerges late (Fig. 3). At the deepest block the symmetric intervention recovers +2.704 and +5.148 nats and mediates 66% and 80% of the trace gap, while the pre-answer control is null or small (Table 6).

This result is robust to the masked span: expanding the re-masked answer by one neighboring token on each side still yields deepest answer-position mediation of 61% (LLaDA) and 75% (Dream), with location-specific effects of 2.272 [1.261, 3.314] and 3.319 [2.649, 4.017] nats, while matched pre-answer controls remain null or small (−0.093 [−0.272, 0.041] and 0.050 [0.018, 0.090]).

Quantity (nats)	LLaDA-8B	Dream-7B
Baseline correct-minus-wrong-trace gap	4.118 [3.378, 4.880]	6.471 [5.473, 7.504]
Deep symmetric patch effect	+2.704 [2.056, 3.407]	+5.148 [4.327, 6.000]
Pre-answer location control	−0.114 [−0.312, 0.036]	+0.044 [0.015, 0.081]
Specific effect over control	+2.818 [2.138, 3.540]	+5.103 [4.290, 5.952]
Fraction of mean trace gap mediated	0.657	0.795

Table 6: Causal patching at the deepest answer-prediction positions. Effects are changes in mean correct-answer log-probability; intervals bootstrap questions.

Repeating the intervention in Qwen and Llama, deep predecessor patches also mediate answer likelihood, but patching an earlier pre-answer span remains strongly causal in the AR models, whereas the diffusion location control is null or small at depth (Table 7). The location-specific diffusion-minus-AR differences are +0.364 [0.212, 0.513] for LLaDA versus Qwen and +0.439 [0.366, 0.506] for Dream versus Llama. The distinction is therefore not that only diffusion output states matter, but that trace-conditioned prediction becomes more causally concentrated at late masked-answer slots in the diffusion models studied.

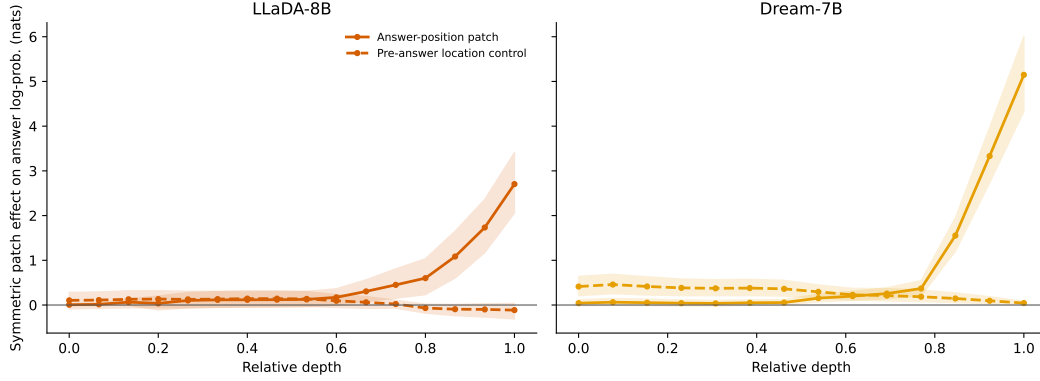


Figure 3: Causal answer-position patching across depth. Solid curves average forward rescue and reverse damage; dashed curves patch a matched pre-answer location. The answer-position effect emerges late in both models. Shading shows 95% question-bootstrap confidence intervals.

Model	Answer-prediction	Pre-answer control	Location-specific
LLaDA-8B	0.657	-0.028	0.684
Dream-7B	0.795	0.007	0.789
Qwen3-8B	0.768	0.447	0.321
Llama-3.1-8B	0.905	0.556	0.349

Table 7: Deep causal effects normalized by each model’s mean trace-conditioned answer-likelihood gap. The location-specific fraction subtracts the pre-answer patch from the answer-position patch. Diffusion rows use 114 LLaDA and 101 Dream questions; AR controls use 60 Qwen and 59 Llama questions.

211 **The effect is dose-dependent and question-specific.** Interpolating the wrong-trace state toward
212 its same-question correct-trace donor raises correct-answer likelihood smoothly (LLaDA rescue
213 $0 \rightarrow 3.813$, Dream $0 \rightarrow 6.302$ nats over $\alpha \in \{0, \dots, 1\}$; positive slope for $> 90\%$ of questions).
214 At full dose the same-question donor beats a donor from another question by $5.170/5.392$ nats, a
215 neuron-shuffled donor by $3.513/5.681$, and the pre-answer patch by $4.100/6.219$, arguing against a
216 generic “high-quality activation” account (Appendix G).

217 **Intermediate trace semantics control compatibility.** Holding the correct answer fixed, changing
218 one intermediate number lowers deep compatibility by $1.108/1.321$ SD and correct-answer likelihood
219 by $1.172/2.052$ nats in LLaDA/Dream, and shuffling reasoning steps lowers them by $1.872/1.571$
220 SD and $3.529/5.663$ nats, whereas an equivalent-format rewrite is far smaller (Fig. 4). Patching the
221 clean late *residual* state into the corrupted run mediates $55\%/78\%$ of the format-adjusted number loss
222 and $97\%/92\%$ of the ordering loss (Table 8). Attention-output patches are positive for both models
223 and corruptions while MLP-output patches disagree in sign, supporting a replicated attention route
224 but not a universal component mechanism. Head-level localization is in Appendix G.

Model	Corruption	Residual patch	Attention patch	MLP patch
LLaDA-8B	Number	0.604 [0.501, 0.714]	0.294 [0.248, 0.341]	-0.065 [-0.086, -0.044]
Dream-7B	Number	1.588 [1.294, 1.915]	0.089 [0.040, 0.140]	0.343 [0.262, 0.433]
LLaDA-8B	Step shuffle	3.350 [2.830, 3.927]	0.854 [0.724, 0.996]	-0.224 [-0.275, -0.171]
Dream-7B	Step shuffle	5.175 [4.248, 6.096]	0.860 [0.679, 1.032]	0.431 [0.330, 0.534]

Table 8: Semantic-corruption mediation at the deepest block, in nats. Each patch effect subtracts the equivalent-format edit and averages clean-to-corrupted rescue with corrupted-to-clean damage. The corresponding format-adjusted likelihood losses are $1.099/2.025$ nats for number corruption and $3.456/5.636$ nats for step shuffling (LLaDA/Dream). The analysis uses 113 LLaDA and 101 Dream questions.

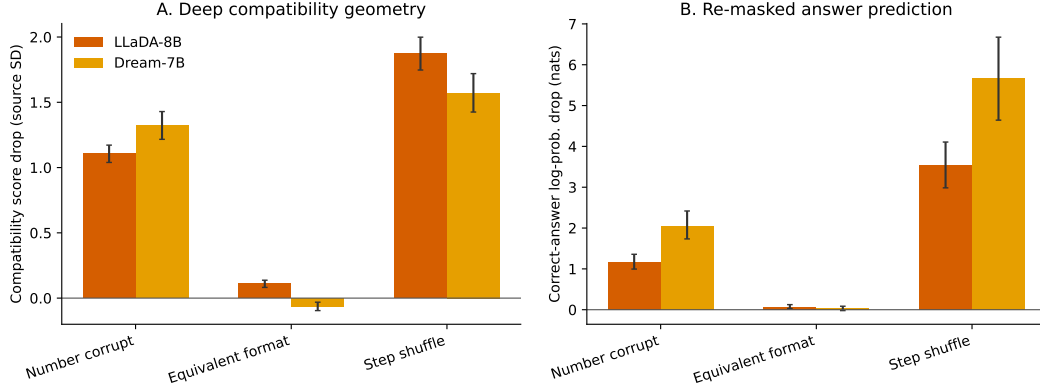


Figure 4: Holding the final answer fixed, intermediate-number corruption and step shuffling reduce both deep compatibility and re-masked correct-answer likelihood; an equivalent-format rewrite has a much smaller effect. Error bars show 95% question-bootstrap confidence intervals.

7 Limitations

Several caveats should be kept in mind when interpreting these results. The GSM8K depth dissociation replicates under a second independent generation seed, but the causal patching and factorial analyses use a single seed, so their question bootstraps capture variation across questions rather than across generation draws or sampling temperatures; robustness of the mediated fractions to generation stochasticity therefore remains untested. The dissociation between depth and gain is strong on GSM8K, but the corresponding result on MATH is same-signed and inconclusive at $n = 150$. The factorial cohorts also vary in size, with 63 Qwen, 79 Llama, 114 LLaDA, and 101 Dream questions. Diffusion patching uses 114 LLaDA and 101 Dream questions, while the component analysis uses 113 and 101. The autoregressive location controls (60 Qwen and 59 Llama) and head analysis (89 LLaDA and 76 Dream) use smaller original cohorts, which limits replication across model families. “Trace source” is an outcome-associated label, not a direct annotation of the reasoning process, so its effect may reflect coherence, style, repetition, or numerical consistency rather than logical validity. Because the pairing requires both a correct and an incorrect generation, it also cannot speak to pass rates or how often mixed outcomes occur naturally. Interpreting the interventions causally has limits: activation transplants can move the model off-manifold, and component patches are not an additive decomposition. Bidirectional rescue/damage, exact self-patch checks, and location controls mitigate but do not eliminate this, so the mediated fractions are intervention effects rather than formal natural indirect effects. The causal experiments re-mask short numeric spans on GSM8K, so longer or non-numeric answers may behave differently and are untested, and the number-corruption, component, and head analyses were not repeated in the AR controls; the replicated attention-output contribution should thus be read as specific to the two diffusion models. Finally, we identify a mediator and output-side contributions but not the source-token pathways or features that construct the representation, nor why it emerges under the diffusion objective—these remain open.

8 Conclusion

LLaDA and Dream form a shared trace-answer compatibility representation late in depth: task- and format-transferable, decodable without correctness labels, aligned across questions, and causally predictive. Patching late answer-prediction states transfers 66% and 80% of the models’ trace-conditioned likelihood gaps; the effect is dose-dependent, needs a question-matched donor, and tracks numerical and ordering corruption far more than an equivalent-format edit, with clean-state restoration mediating 55–78% of the number loss and 92–97% of the ordering loss. AR controls show compatibility is not diffusion-unique at the endpoint; the diffusion-specific finding is that it is weak shallowly, organizes at depth, and concentrates causally at masked-answer slots. This moves latent correctness analysis from label decoding to what is represented and where it acts, leaving open the source-token paths that build the state.

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A Measurement Details

For question i and sampled generation k , let $c_{ik} \in \{0, 1\}$ denote exact-match correctness. A readout at transformer block ℓ with pooling rule p produces

$$s_{ik}^{(\ell,p)} = f_{\ell,p}\left(h_{ik}^{(\ell)}\right),$$

where $f_{\ell,p}$ is an L2-regularized logistic regression ($C = 0.1$) over standardized states. Probes are fitted on training folds and evaluated out-of-fold, with all K generations of a question assigned to the same fold.

Pooled AUC compares every correct generation with every incorrect generation:

$$A_{\text{pool}} = \frac{\sum_{i,k:c_{ik}=1} \sum_{j,m:c_{jm}=0} \phi(s_{ik}, s_{jm})}{N_+ N_-}, \quad \phi(a, b) = \mathbb{I}[a > b] + \frac{1}{2}\mathbb{I}[a = b].$$

Most comparisons have $i \neq j$. A readout that only distinguishes easy from hard questions can therefore score highly without ranking attempts to the same question.

Within-question AUC instead restricts the comparison to correct–incorrect pairs from the same question:

$$A_{\text{within}} = \frac{\sum_i \sum_{k:c_{ik}=1} \sum_{m:c_{im}=0} \phi(s_{ik}, s_{im})}{\sum_i n_i^+ n_i^-}.$$

Questions whose K generations all have the same label contribute no pairs. This micro-average weights each observed within-question pair equally.

For all-position pooling, depth gain is

$$\Delta_{\text{depth}} = A_{\text{within}}^{(\ell_D, \text{all})} - A_{\text{within}}^{(\ell_0, \text{all})},$$

with both endpoints fixed in advance. Here ℓ_0 is block 0 and ℓ_D is block 30 of 32 for LLaDA, block 34 of 36 for Qwen, and block 26 of 28 for Dream. Confidence intervals resample questions 2,000 times while keeping their generations together and holding the out-of-fold probe scores fixed. They capture evaluation-set variation, but not uncertainty from probe refitting, fold reassignment, model checkpoints, or a new generation seed.

B Within-Question Validation

Across 36 settings spanning AR and diffusion models, GSM8K, MATH, and BIG-Bench-Hard, multiple temperatures, diffusion block sizes, and values of K , within-question AUC is the strongest predictor of probe-guided best-of- K selection gain. Its correlation with gain is $r = 0.762$ ($p = 6.57 \times 10^{-8}$), compared with $r = 0.525$ for the between-question signal and $r = 0.500$ for pooled AUC.

Leave-one-setting-out prediction from within-question AUC reaches $R^2 = 0.537$ with MAE 0.030, versus $R^2 = 0.171$ and MAE 0.040 for pooled AUC. A relation fitted only on AR settings predicts diffusion settings with MAE 0.023. Diffusion settings therefore occupy a lower region of the same empirical relation rather than requiring a separate mapping.

Readout statistic	Correlation with gain	LOO R^2	Interpretation
Within-question AUC	0.762	0.537	ranks attempts
Between-question signal	0.525	0.184	predicts difficulty
Pooled AUC	0.500	0.171	mixes both
Mean-only baseline	–	–0.058	no setting information

Table 9: Correctness-readout statistics versus probe-selection gain across all 36 settings.

The distinction is visible in individual settings. Under the original intermediate-layer GSM8K protocol, Qwen has pooled AUC approximately 0.95 and within-question AUC approximately 0.91, whereas LLaDA has pooled AUC approximately 0.69 but within-question AUC approximately 0.62. Thus a pooled readout can appear useful while remaining weak at choosing among attempts to one prompt. This helps explain why high latent AUROC need not produce a strong best-of- N selector.

C Surface and First-Block Floors

Seven features of the parsed answer string, with no access to model internals or the reasoning trace, already reach within-question AUC 0.780 for Qwen, 0.634 for LLaDA, and 0.673 for Dream. Answer form therefore carries correctness information even after the question is held fixed.

The first-block hidden-state floor differs sharply by architecture (Table 10). Qwen’s terminal-bin readout reaches 0.913, whereas LLaDA is at chance at 0.492 and Dream is much lower at 0.529. The Qwen result is not explained by length: restricting to correct–incorrect pairs with exactly equal generated length retains AUC 0.890. Nor is the LLaDA result caused by averaging over filler: its exact-answer first-block AUC is 0.488, and no tested shallow answer-region variant exceeds approximately 0.538. These measurements are interpretive floors, not quantities to subtract from deeper AUCs.

Model	Answer only	First/all	First/terminal	First/exact answer	Deepest/all
Qwen3-8B	0.780	0.820	0.913	0.824	0.869
LLaDA-8B	0.634	0.451	0.492	0.488	0.826
Dream-7B	0.673	0.556	0.529	0.529	0.800

Table 10: Within-question AUC under progressively richer GSM8K readouts. “First” is the output of transformer block 0, not the raw embedding; “terminal” is the final relative-position bin; and “exact answer” uses detected answer-token positions. Dream’s terminal and exact-answer estimates independently round to 0.529.

D Objective-Aligned Answer Scoring

For diffusion models, we jointly re-mask the generated numeric answer span and score the mean conditional log probability assigned to its original tokens. For AR models, we use the teacher-forced mean next-token log probability over the same answer span. This produces a training-free scalar with the natural orientation that higher likelihood indicates greater correctness.

On GSM8K, the LLaDA score reaches within-question AUC 0.730 [0.654, 0.800]. Its interval is disjoint from Qwen at 0.483 [0.449, 0.519] and Llama at 0.394 [0.324, 0.460]. Dream reaches 0.668. This is a measurement dissociation induced by differing predictive objectives, not evidence that diffusion models intrinsically know when they are correct.

Model	Class	Decode-confidence AUC	Conditional answer AUC	95% CI
LLaDA-8B	diffusion	0.639	0.730	[0.654, 0.800]
Dream-7B	diffusion	0.658	0.668	–
Qwen3-8B	AR	0.566	0.483	[0.449, 0.519]
Llama-3.1-8B	AR	0.559	0.394	[0.324, 0.460]

Table 11: Within-question AUC of training-free confidence scores on GSM8K. Diffusion likelihood jointly re-masks the answer span; AR likelihood scores that span left-to-right.

Re-masking does not improve an ordinary hidden-state correctness probe. LLaDA block-0 AUC changes from 0.488 to 0.487 and its deepest AUC from 0.831 to 0.835; Dream changes from 0.529 to 0.536 at block 0. The useful conditional-likelihood scalar is therefore not evidence that masking alone localizes a better hidden-state correctness representation.

E Relation to Prior Readout Protocols

Yadav et al. [2026] use one generation per prompt, which supports pooled AUROC but not within-question AUC. They also report a prompt-only score of approximately 0.647 relative to a latent readout around 0.720, explicitly showing that question difficulty explains substantial signal.

We instantiate a protocol bridge in our own harness: terminal-bin pooling, layer selection by pooled out-of-fold AUROC, and repeated draws of one of the six candidates per question. The one-sample

363 pooled values closely track the six-sample pooled values because selecting one candidate removes
 364 within-question ordering information but preserves between-question difficulty.

Model	Selected block	Pooled	One-sample pooled	Same-state within	Best all-position within
LLaDA-8B	30	0.686	0.684 ± 0.032	0.776	0.826
Qwen3-8B	26	0.967	0.967 ± 0.012	0.930	0.900
Dream-7B	20	0.777	0.776 ± 0.025	0.706	0.800

Table 12: Protocol bridge. “Pooled” uses all six out-of-fold scores; “one-sample pooled” averages AUROC over random draws of one generation per question. The last two columns expose the effect of changing the target metric or pooling rule.

365 The bridge exposes two facts hidden by a single pooled number. First, Qwen’s block-0 terminal
 366 state already has pooled AUROC 0.939 and within-question AUC 0.913. Second, pooled and within-
 367 question layer selection can disagree: Dream’s pooled-selected state has within-question AUC 0.706,
 368 whereas deepest all-position pooling reaches 0.800. Pooled AUROC is therefore not incorrect; it
 369 simply cannot identify whether a readout uses question difficulty, answer form, or attempt-specific
 370 internal information.

371 F Mechanistic Tests and Negative Results

372 None of five broad explanations accounts for the depth-formation pattern. **Parallel decoding:** at fixed
 373 compute, LLaDA block sizes 256/64/32/8 give within AUC 0.664/0.784/0.720/0.704, uncorrelated
 374 with AR-likeness ($r = 0.22$, $p = 0.78$). **Commit order:** reordering states by denoising commit step
 375 gives a positive LLaDA slope (+0.125) but a negative Dream slope (-0.022), so it is not a general
 376 hidden aggregation axis. **Loss-support:** re-masking does not improve hidden-state readouts (App. D).
 377 **Probe depth vs. causal importance:** single-layer ablations show similar early/terminal sensitivity
 378 across model classes rather than the depth-gain shift, so accessibility and causal necessity differ.
 379 **Selection:** combining probe score, agreement, and confidence beats majority voting by only 0.011
 380 ($p = 0.14$), and probe-guided allocation matches random. The paper is therefore an interpretability
 381 study with enabling controls, not a new selector.

382 G Additional Results

383 G.1 Reader-capacity ladder

384 Increasing reader capacity under identical folds does not erase the raw accessibility gap (Ta-
 385 bles 13, 14); per-token and learned pooling help LLaDA, but its best reader stays below Qwen’s
 386 weakest, so the gap is not solely a weak-linear-probe artifact.

Mean-pooled reader	Qwen3-8B	LLaDA-8B	Dream-7B
Linear, one layer	0.905	0.615	0.766
Linear, all-layer concat.	0.903	0.623	0.768
MLP, all-layer concat.	0.881	0.652	0.685
Gradient boosting	0.838	0.639	0.719
Linear, per-question centered	0.892	0.613	0.728

Table 13: Within-question AUC for a mean-pooled capacity ladder.

387 G.2 Further controls

388 Prompt-only difficulty explains most of LLaDA’s between-question signal (best prompt-only correla-
 389 tion 0.412 vs. generation 0.438; AR 0.651 vs. 0.862). Difficulty-matched within-question AUC is
 390 0.903 (Qwen) vs. 0.625 (LLaDA), with LLaDA weakest on single-correct questions (0.489 at $j=1$
 391 rising to 0.685 at $j=5$). Majority voting beats the original LLaDA probe on numeric GSM8K (0.670
 392 vs. 0.625) but the probe wins on all four examined BIG-Bench-Hard settings. About 2.5%/2.6% of

Per-token reader	Qwen3-8B	LLaDA-8B
Mean pool + linear	0.899	0.658
Mean pool + MLP	0.901	0.666
Max pool + linear	0.947	0.734
Attention pool + MLP	0.941	0.739
Last token + linear	0.956	0.657

Table 14: Within-question AUC for readers over per-token states.

AR/diffusion generations lack a parseable answer; restricting to fully-parseable questions preserves the probe’s advantage over decode confidence.

H Reproducibility

All generation and analysis were run in Kaggle GPU notebooks (NVIDIA T4/P100), with models loaded from the Hugging Face hub. The primary GSM8K and MATH artifacts are complete at 150 questions with $K = 6$ generations for all four models; a partial second seed is excluded. The crossed cohorts contain 114 LLaDA, 101 Dream, 63 Qwen, and 79 Llama questions, with all twelve variants per question passing numeric checks. Diffusion activation patching and donor-specificity analyses use 114 LLaDA and 101 Dream questions. The semantic-component analysis uses 113 LLaDA and 101 Dream questions after excluding one LLaDA question without a valid semantic corruption. The objective-matched AR location controls remain on 60 Qwen and 59 Llama questions, while the individual-head analysis remains on the original 89 LLaDA and 76 Dream questions.

All-correct and all-wrong prompts are resampled until the missing outcome appears, with up to 48 additional attempts. This targeted sampling is used only for paired interventions and never for pass-rate estimation. Exact self-patches reproduce the unpatched score to machine precision. Generators checkpoint their progress, and each analysis truncates to the completion marker, so no partial artifact is treated as complete.

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